BTP Research Presentation

Indian Institute of Information Technology Kottayam

April 10, 2026

1. Motivation
2. Problem and Data
3. Baselines and Experiments
4. Graph-Aware Model
5. Results and Interpretation
6. Graph Algorithms
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The original system already had:

- Strong pairwise stance signals from transformer models
- Graph construction and graph algorithms for debate analysis

The core bottleneck was that the graph analysis was only as good as the stance predictions. Our central research question is:

Research Question

Can we improve stance detection by incorporating debate-local graph structure rather than classifying each claim pair in isolation?

The older presentation already identified the central limitation:

- The graph modelling and graph algorithms were in place
- The overall system still struggled on complex debates
- The main bottleneck was the stance prediction module

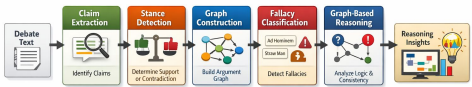
This motivates a progression of experiments:

1. Direct NLI baseline
2. Direct stance baseline
3. Fused DeBERTa + stance logistic regression baseline
4. Graph-aware model with debate-local semantic structure

The previous framework treated the problem as a modular pipeline:

- Claim extraction
- Stance detection
- Graph construction
- Graph analysis
- Contradiction and cycle reasoning

The new work preserves this structure but focuses on improving the stance component so that downstream graph reasoning becomes more reliable.



Each candidate edge is a directed relation:

$$e = (u, v, y)$$

where:

- u: source claim
- v: target claim
- $y \in \{\texttt{support}, \texttt{attack}, \texttt{neutral}\}$

The textual convention used throughout the experiments is:

- Premise = target claim text
- Hypothesis = source claim text

Dataset used for benchmarking:

- SyncIALO, a synthetic Kialo-style debate graph dataset
- Corpus: `synthetic_corpus-001`
- Split used in the saved benchmark: `eval`

Saved benchmark configuration:

- Debates used: 8
- Train debates: 6
- Test debates: 2
- Training examples: 7224
- Test examples: 2664



(a) An example of the arguments made in a Kialo debate. The thesis is, "*Pregnant people should have the right to choose abortion*". A supporting reply is, "*Access to legal abortion improves the health and safety of women*". An attacking reply to this support is, "*When abortion is easily available, it incentivises irresponsible behaviour*".

Split	Support	Attack	Neutral	Total
Train	1806	1806	3612	7224
Test	666	666	1332	2664
Overall	2472	2472	4944	9888

Observation:

- The benchmark is balanced between support and attack
- Neutral is intentionally larger because sampled non-edges are included

We benchmark four increasingly stronger models:

1. Old DeBERTa rule/decision baseline
2. Old stance-only baseline
3. DeBERTa + stance logistic regression baseline
4. Graph-aware Random Forest model

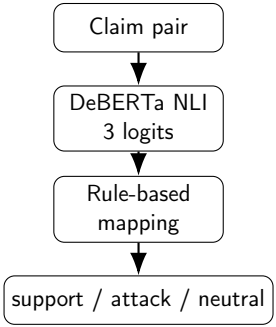
This sequence lets us isolate where performance gains come from:

- Stronger base model choice
- Model fusion
- Graph-aware local context

- Input: source-target claim pair
- Model: DeBERTa NLI
- Output logits: contradiction, neutral, entailment
- Final label obtained through an older rule-based decision layer

Key weakness:

- Rich textual inference, but no stance-specific training and no graph context



Accuracy: 0.5601, Macro-F1: 0.3907, Weighted-F1: 0.4658

Class	Precision	Recall	F1
Support	0.8182	0.1081	0.1910
Attack	0.8722	0.1742	0.2904
Neutral	0.5338	0.9790	0.6909

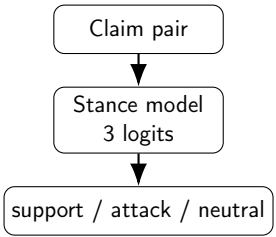
Interpretation:

- The model is extremely conservative for support and attack
- It overpredicts neutral, giving high neutral recall
- NLI evidence alone is not sufficient for discourse stance classification

- Input: source-target claim pair
- Model: target-dependent stance classifier
- Output logits: against, none, favor
- Final relation predicted directly from stance outputs

Key weakness:

- Stance specialization helps some classes
- But there is still no NLI fusion and no graph context



Accuracy: 0.3517, Macro-F1: 0.3104, Weighted-F1: 0.3349

Class	Precision	Recall	F1
Support	0.2607	0.7162	0.3822
Attack	0.2844	0.0931	0.1403
Neutral	0.6461	0.2988	0.4086

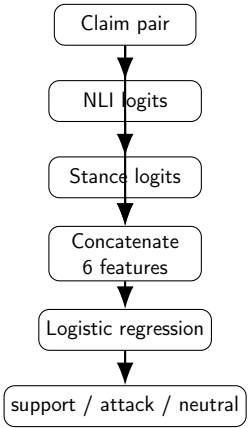
Interpretation:

- The stance model overpredicts support
- Support recall improves, but attack performance collapses
- The two single-model baselines fail in opposite directions

The next idea is feature fusion:

- Concatenate NLI logits and stance logits
- Create a 6-dimensional fused feature vector
- Train a multinomial logistic regression classifier

This baseline tests whether the two transformer models are complementary.



Accuracy: 0.6513, Macro-F1: 0.6557, Weighted-F1: 0.6477

Class	Precision	Recall	F1
Support	0.5726	0.7102	0.6340
Attack	0.6257	0.8183	0.7092
Neutral	0.7415	0.5383	0.6237

Interpretation:

- This is the first genuinely balanced baseline
- Fusion successfully offsets the single-model biases
- However, classification is still pairwise and local

The logistic-regression baseline is much stronger, but it still ignores debate-local structure. Questions it cannot answer directly:

- Is the source claim central or isolated in the debate?
- Does the target lie inside the source's semantic neighborhood?
- Do the two claims share the same semantic cluster?
- Is the pair locally embedded in the same debate subtopic?

The graph-aware model adds exactly this missing context.

Graph features do not replace pairwise semantics. They solve a different problem.

Key Intuition

Pairwise models answer: "What do these two claims seem to say to each other?"
Graph features answer: "Where do these two claims sit inside the local debate neighborhood?"

Why this matters:

- Two claim pairs can look textually similar but play different discourse roles.
- Topical similarity alone does not tell us whether the relation is support, attack, or just same-topic neutral.
- Local neighborhood structure tells us whether the two claims belong to the same argumentative region, whether one is central, and whether the pair is locally important.

Consider three claims in the same debate:

- Claim A: "Governments should subsidize renewable energy."
- Claim B: "Public funding is necessary to speed up the energy transition."
- Claim C: "Subsidies distort markets and reduce competition."

A pairwise model sees topic overlap in all three claims because they all discuss subsidies and energy policy.

The graph-aware view adds a clearer interpretation:

- A and B are likely to share neighbors and appear in the same support cluster.
- C may still be topically close, but it will often sit in a different neighborhood pattern and connect differently to nearby claims.
- Structure therefore helps separate "same topic" from "same stance".

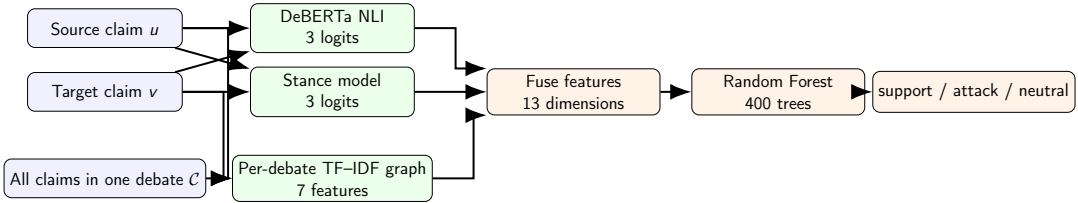
The graph-aware model combines:

1. Pairwise NLI evidence
2. Pairwise stance evidence
3. Debate-local semantic graph features

This yields:

$$x_{\text{full}} = \left[x_{\text{NLI}} \; ; \; x_{\text{stance}} \; ; \; x_{\text{graph}}\right] \in \mathbb{R}^{13}$$

The final classifier is a Random Forest with 400 trees.


Source claim u
Target claim v
All claims in one debate $\mathcal{C}$
DeBERTa NLI
3 logits
Stance model
3 logits
Per-debate TF–IDF graph
7 features
Fuse features
13 dimensions
Random Forest
400 trees
support / attack / neutral

For each debate, TF–IDF is fit *only on claims in that debate*.

$$\mathrm{tf}(t, c_i) = \text{count of term } t \text{ in claim } c_i$$

$$\mathrm{idf}(t, \mathcal{C}) = \log\left(\frac{1 + |\mathcal{C}|}{1 + |\{c \in \mathcal{C} : t \in c\}|}\right) + 1$$

$$\mathrm{tfidf}(t, c_i, \mathcal{C}) = \mathrm{tf}(t, c_i) \cdot \mathrm{idf}(t, \mathcal{C})$$

Why per debate?

- Topics are local to each debate.
- Salience should be measured within the current debate, not globally.
- This makes the graph features debate-sensitive rather than corpus-generic.

Let $\mathbf{t}_i$ be the TF–IDF vector of claim c_i. Similarity is:

$$\mathrm{sim}(c_i, c_j) = \frac{\mathbf{t}_i^\top \mathbf{t}_j}{\|\mathbf{t}_i\|_2 \|\mathbf{t}_j\|_2}$$

For each claim, retain neighbors satisfying:

- Cosine similarity $\geq \tau$.
- Membership among the top-k neighbors.

Saved benchmark values:

$$k = 8, \qquad \tau = 0.15$$

This produces the local neighborhood set:

$$N_k(c_i)$$

1. Cosine similarity: $f_1(u, v) = \mathrm{sim}(u, v)$
2. Normalized source degree: $f_2(u, v) = \frac{|N_k(u)|}{\max(1,k)}$
3. Normalized target degree: $f_3(u, v) = \frac{|N_k(v)|}{\max(1,k)}$
4. Target-in-source: $f_4(u, v) = 1[v \in N_k(u)]$
5. Source-in-target: $f_5(u, v) = 1[u \in N_k(v)]$
6. Neighbor-set Jaccard overlap:
$$f_6(u, v) = \frac{|N_k(u) \cap N_k(v)|}{|N_k(u) \cup N_k(v)|}$$
7. Token-level Jaccard overlap:
$$f_7(u, v) = \frac{|T(u) \cap T(v)|}{|T(u) \cup T(v)|}$$

1. Normalized source degree

$$f_2(u, v) = \frac{|N_k(u)|}{\max(1, k)}$$

What problem it solves:

- A claim can be very central or very isolated in the debate.
- Pairwise models cannot tell whether the source is broadly connected to many nearby claims.

Intuition:

- High source degree often means the source is semantically broad or central.
- Low source degree often means the source is very specific or isolated.

Example:

- If claim u is similar to many claims about "renewable energy policy", it is probably a central debate claim rather than a fringe statement.

2. Normalized target degree

$$f_3(u, v) = \frac{|N_k(v)|}{\max(1, k)}$$

This solves the same problem for the target. It helps the model know whether the target is a core motion claim or a peripheral claim.

3. Target-in-source

$$f_4(u, v) = 1[v \in N_k(u)]$$

What problem it solves:

- Cosine similarity alone tells us how similar two claims are.
- It does not tell us whether that similarity is strong enough to make the target one of the source's *important local neighbors*.

Example:

- If u says "Subsidize renewables now" and v says "Government should fund clean energy", then v may be one of the strongest semantic anchors for u.

4. Source-in-target

$$f_5(u, v) = 1[u \in N_k(v)]$$

Why it matters:

- Neighborhood structure is directed.
- v may be a strong neighbor of u, but u may not be a strong neighbor of v.
- This asymmetry captures specificity vs generality.

5. Neighbor-set Jaccard overlap

$$f_6(u, v) = \frac{|N_k(u) \cap N_k(v)|}{|N_k(u) \cup N_k(v)|}$$

What problem it solves:

- Two claims may not directly be nearest neighbors.
- They may still live in the same local semantic region of the debate.

Example:

- One claim talks about emissions, another about renewable subsidies.
- If both connect to the same cluster of climate-policy claims, they likely belong to the same argumentative neighborhood.

6. Token-level Jaccard overlap

$$f_7(u, v) = \frac{|T(u) \cap T(v)|}{|T(u) \cup T(v)|}$$

What problem it solves:

- TF–IDF can miss simple literal reuse patterns.
- Token overlap recovers cases where two claims share key surface words or phrases.

Each graph feature solves a different weakness of pure pairwise prediction:

- **Degree features** tell us whether the source and target are central or peripheral.
- **Membership features** tell us whether the pair is locally important in each other's neighborhood.
- **Overlap features** tell us whether the two claims belong to the same local semantic region.

Why this is better than just cosine similarity:

- Cosine similarity is only one local number.
- The debate graph needs structure: centrality, asymmetry, and neighborhood overlap.
- These features let the model reason about context without building a full GNN.

This is why the graph-aware model is not just "pairwise model + one similarity score". It is a structured local-context model.

The final classifier receives a 13-dimensional fused representation.
Random Forest is used because:

- The 6 transformer logits are useful but not fully linear.
- The graph features create thresholded and interaction-heavy patterns.
- Tree ensembles can capture nonlinear combinations naturally.

Prediction rule:

$$\hat{p}(x) = \frac{1}{M} \sum_{m=1}^{M} T_m(x), \qquad M = 400$$

$$\hat{y} = \arg\max_c \hat{p}_c(x)$$

1. Load debate graphs.
2. Parse gold support and attack edges.
3. Sample neutral non-edges.
4. Split debates into train and test by debate, not by row.
5. Compute 6 transformer features.
6. Compute 7 graph features within each debate.
7. Concatenate into 13-dimensional vectors.
8. Train Random Forest on train debates only.
9. Evaluate on held-out debates.

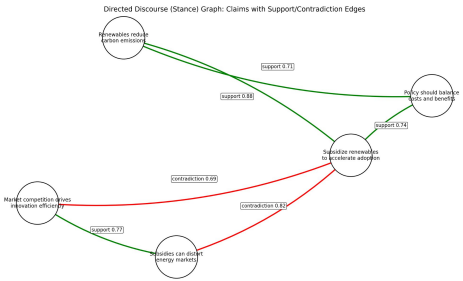
Suppose we want to classify the edge from:

- Source: “Subsidize renewables to accelerate adoption”
- Target: “Policy should balance costs and benefits”

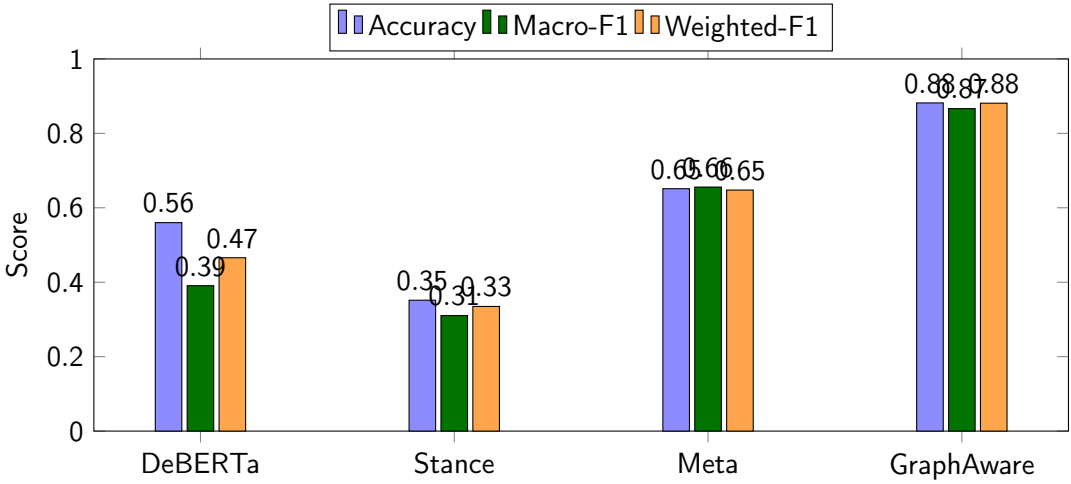
A pairwise model only sees the two sentences.
A graph-aware model also asks:

- Is the source central in the renewable-energy cluster?
- Is the target one of the source’s closest semantic anchors?
- Do they share the same climate-policy neighborhood?

That extra context makes support vs attack vs neutral much easier.



Model	Accuracy	Macro-F1	Weighted-F1
DeBERTa baseline	0.5601	0.3907	0.4658
Stance baseline	0.3517	0.3104	0.3349
DeBERTa + stance LR	0.6513	0.6557	0.6477
Graph-aware	**0.8818**	**0.8665**	**0.8811**


Accuracy
Macro-F1
Weighted-F1
Score
1
0.8
0.6
0.4
0.2
0
0.56
0.39
0.47
0.35
0.31
0.33
0.65
0.66
0.65
0.88
0.87
0.88
DeBERTa
Stance
Meta
GraphAware

	precision	recall	f1-score
support	0.9677	0.0617	0.1161
attack	0.2672	0.9374	0.4158
neutral	0.0682	0.0120	0.0204
accuracy			0.2605
macro avg	0.4344	0.3370	0.1841
weighted avg	0.3376	0.2605	0.1446

DeBERTa rule baseline

	precision	recall	f1-score
support	0.2439	0.5329	0.3346
attack	0.2756	0.1526	0.1965
neutral	0.5710	0.3729	0.4511
accuracy			0.3555
macro avg	0.3635	0.3528	0.3274
weighted avg	0.4160	0.3555	0.3578

stance-only baseline

Accuracy: 0.6252072968490879

```
[[218  17  62]
 [ 20 237  44]
 [155 154 299]]
```

	precision	recall	f1-score
support	0.5547	0.7340	0.6319
attack	0.5809	0.7874	0.6685
neutral	0.7383	0.4918	0.5903
accuracy			0.6252
macro avg	0.6246	0.6711	0.6303
weighted avg	0.6538	0.6252	0.6201

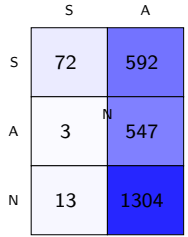
fused DeBERTa + stance LR

Class	Precision	Recall	F1
Support	0.8485	0.8243	0.8363
Attack	0.8525	0.8243	0.8382
Neutral	0.9111	0.9392	0.9250

Interpretation:

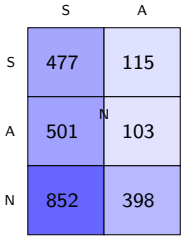
- Performance is strong across all three classes.
- Support and attack are no longer badly imbalanced.
- Neutral remains strong without overwhelming the other two classes.

DeBERTa baseline



Strong neutral bias. Most support and attack examples collapse into neutral.

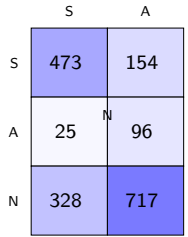
Stance baseline



Strong support bias. The model pulls many attack and neutral examples toward support.

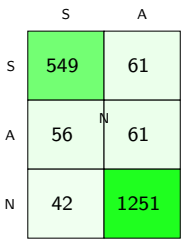
Rows are gold labels and columns are predicted labels. Abbreviations: S = Support, A = Attack, N = Neutral.

DeBERTa + stance LR



Better balance than the single-model baselines, but many neutral cases are still confused with support and attack.

Graph-aware



Clean diagonal across all three classes. Structural context reduces both neutral confusion and support-attack cross-talk.

Rows are gold labels and columns are predicted labels. Darker cells indicate larger counts.

Why does the graph-aware model help so much?

1. Debate-local context reduces ambiguity.
2. Graph features separate true neutral non-edges more effectively.
3. Semantic neighborhood overlap adds structure not available to pairwise models.
4. Nonlinear interactions between transformer evidence and graph signals matter.

Compared to the DeBERTa + stance logistic regression baseline:

$$\Delta\text{Accuracy} \approx 0.2305$$

$$\Delta\text{Macro-F1} \approx 0.2108$$

Once stance edges are reliable, the graph becomes meaningful enough for discourse analysis. These algorithms answer different questions:

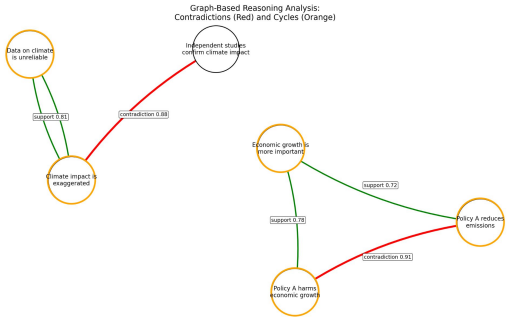
- Which claims are globally well-supported?
- Which claims are globally attacked?
- Which nodes are bottlenecks in the support structure?
- Which claims form coherent topical clusters?
- Where do contradiction paths and cycles appear?

Goal:

- Find explicit contradiction edges.
- Find directed cycles that indicate circular reasoning.

Why useful:

- Contradiction edges expose direct conflict.
- Cycles expose reasoning loops across multiple claims.



Idea:

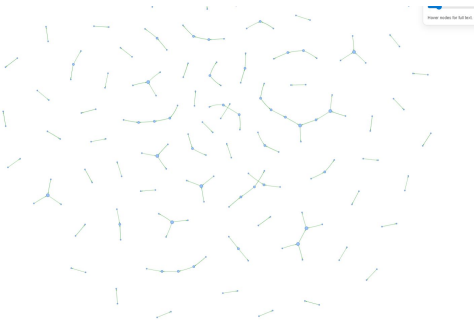
- Count how many support edges point into each claim.

What problem it solves:

- Gives a simple measure of local support strength.
- Helps identify weakly supported claims.

Limitation:

- Only local, and does not know whether the supporting claims are themselves important.



Goal:

- Rank claims by global support influence.

Logic:

- A claim is important if it is supported by other important claims.

Why better than in-degree:

- Not all supporting claims should count equally.

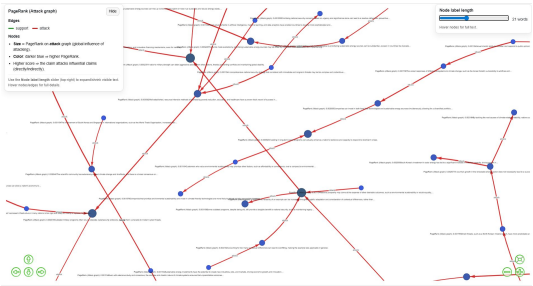


Goal:

- Rank claims by global attack influence.

Interpretation:

- A node with high attack PageRank is attacked by influential attack pathways.
- This helps identify globally controversial or vulnerable claims.



We compute:

$$\text{Net Influence}(u) = PR_{\text{support}}(u) - PR_{\text{attack}}(u)$$

Why this helps:

- Support and attack can both be globally strong.
- This score summarizes whether a claim is overall defended or challenged.

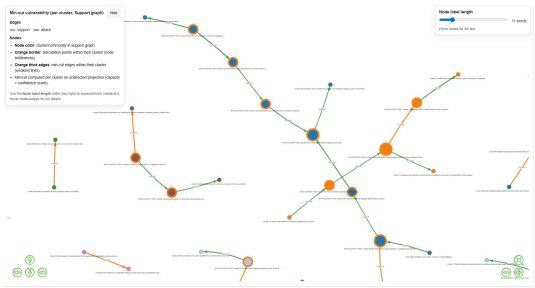


Goal:

- Find bottleneck claims and weak links inside a support structure.

Interpretation:

- If a small cut can break a support cluster, the argument is structurally fragile.
- This is useful for finding vulnerable subarguments.

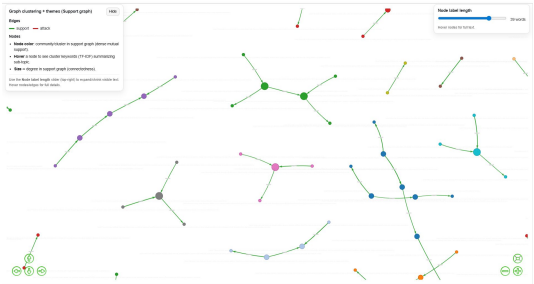


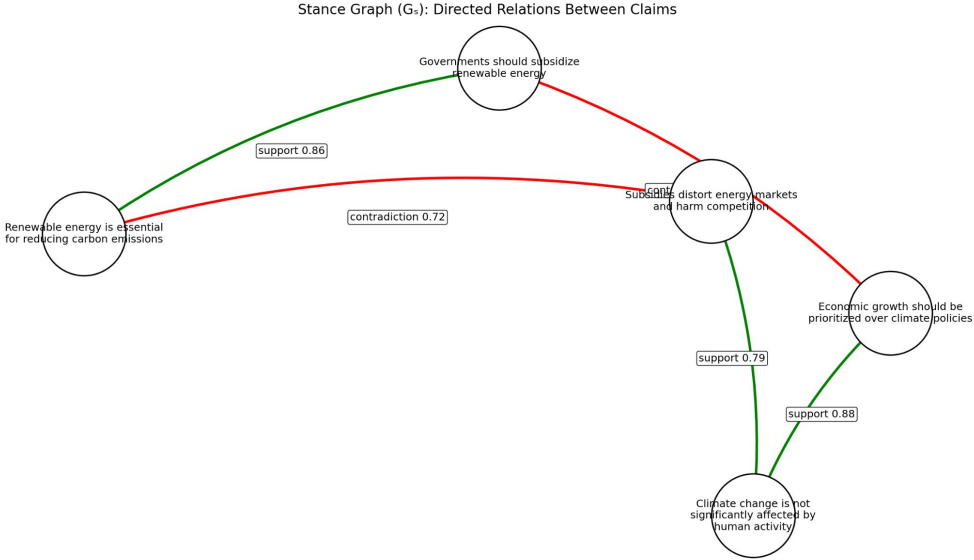
Goal:

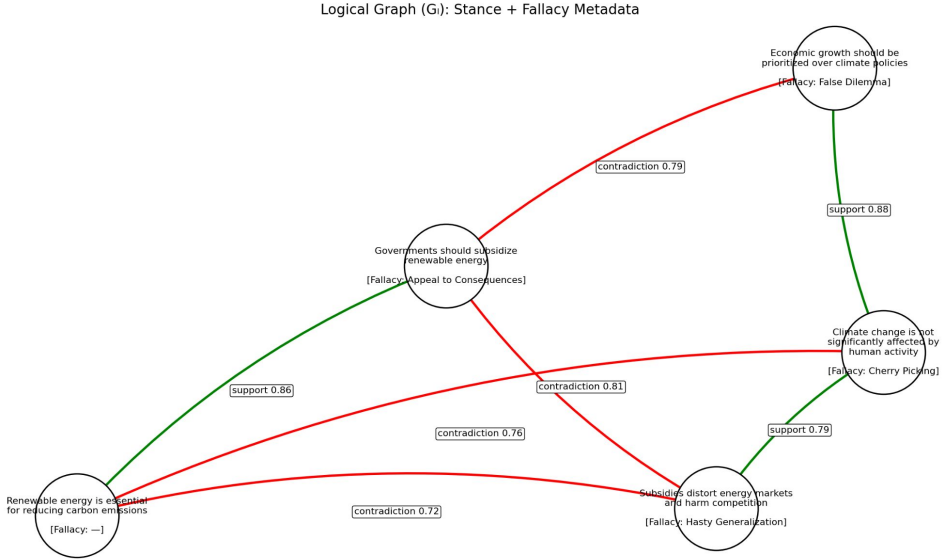
- Group claims into coherent support communities.

What it solves:

- Large debates often contain multiple subtopics.
- Clustering helps reveal local themes and debate substructures.




Stance Graph (G_s): Directed Relations Between Claims
Governments should subsidize renewable energy
support 0.86
Renewable energy is essential for reducing carbon emissions
contradiction 0.72
Subsidies distort energy markets and harm competition
Economic growth should be prioritized over climate policies
support 0.79
support 0.88
Climate change is not significantly affected by human activity


Logical Graph (Gₗ): Stance + Fallacy Metadata
Economic growth should be prioritized over climate policies
[Fallacy: False Dilemma]
contradiction 0.79
support 0.88
Governments should subsidize renewable energy
[Fallacy: Appeal to Consequences]
Climate change is not significantly affected by human activity
[Fallacy: Cherry Picking]
support 0.86
contradiction 0.81
contradiction 0.76
support 0.79
Renewable energy is essential for reducing carbon emissions
[Fallacy: —]
contradiction 0.72
Subsidies distort energy markets and harm competition
[Fallacy: Hasty Generalization]

The earlier presentation already showed that downstream graph analysis depends on good edges:

- Contradiction detection
- Cycle detection
- Support weakness analysis
- Clustering and discourse evolution

The graph-aware model improves these downstream stages because the graph itself becomes less noisy and more semantically coherent.

The earlier work also emphasized semantic pruning and scalability.
This fits naturally with the graph-aware view:

- Not all claim pairs should be treated equally.
- Local semantic neighborhoods provide a principled way to restrict attention.
- Graph-aware stance prediction is therefore also a scalability-aware design direction.

DebateIdx	VertexCount	EdgeCount	OldTime(s)	NewTime(s)	Speedup
0	15	210	186.49	15.62	11.94×
1	27	992	976.67	117.62	8.30×
2	26	650	746.06	84.55	8.82×
3	16	272	142.84	19.10	7.48×
4	34	1122	503.69	76.14	6.61×

- The main weakness of the old system was stance prediction, not graph reasoning itself.
- Direct NLI and direct stance baselines fail in opposite ways.
- DeBERTa + stance logistic regression is a strong non-graph baseline.
- The graph-aware model adds debate-local semantic structure and achieves the best results by a large margin.

- Benchmark on more debates and multiple random seeds.
- Perform ablations over the 7 graph features.
- Compare TF–IDF graph features to embedding-based graph features.
- Move toward iterative graph refinement or a true graph neural model.

Thank You